

Geometry
Unit 12
Lesson 2 Practice

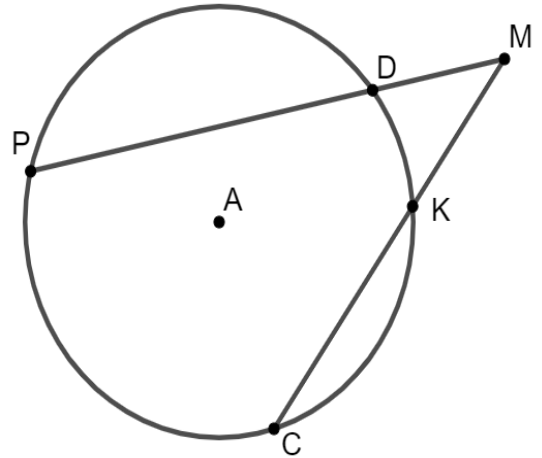
Name _____
Date _____
Period _____

Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = 17$, $DM = 3$, $CK = x$, and $KM = 5$.

1. What is the length of secant $\overline{PM}$?

2. Find the value of x .

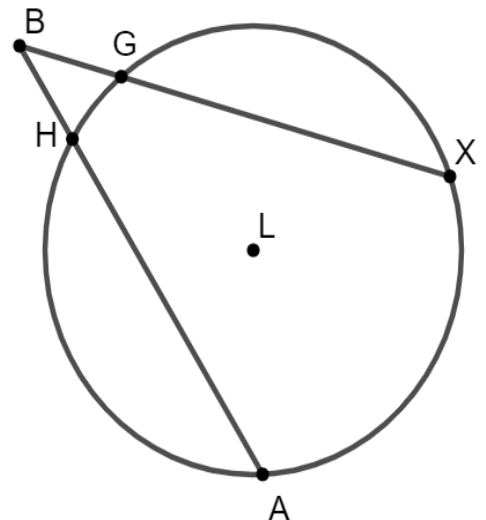
3. What is the length of secant $\overline{CM}$?



Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 8$, $GX = (7x - 4)$, $BH = 6$, and $HA = 18$.

4. What is the value of x ?

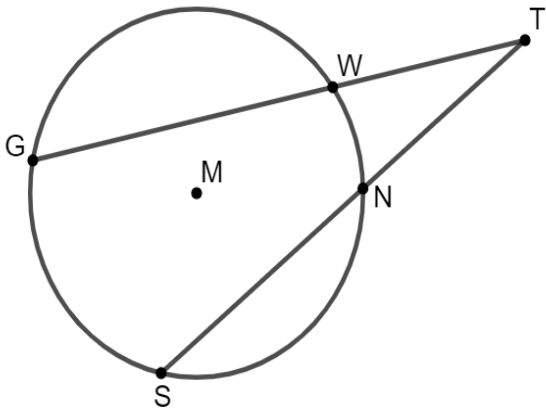
5. Calculate the length of secant $\overline{BX}$.



Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 7x$, $WT = 2x$, $SN = 6$, and $NT = 6$.

6. Find the value of x .

7. What is the length of secant $\overline{GT}$?



Bailey and Luciano were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 2$, $JM = (12x - 5)$, $PK = 3$, and $KR = 4x$.

Each of their equations are set up to solve for x .

Bailey's Work	Luciano's Work
$3(4x + 3) = 2(12x - 3)$	$4x(4x + 3) = 2(12x - 3)$

8. Who wrote the correct equation needed to solve for x ?

9. Use this equation to find the value of x .

10. What is the length of secant $\overline{PR}$?

